

Technical Note Number 8

Users' Manual for the Reconstruction
Package REPACK

by

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ABSTRACT

Multivariate canonical analysis is used to specify tree-growth and climate relationships and to reconstruct anomalies in past climate from anomalies in tree-ring width. This is done by a set of computer programs which was first developed to perform the necessary canonical correlation, canonical regression, and reconstruction of sea level pressure anomalies (predictands) from tree growth anomalies (predictors). The programs were built into a jobstream and generalized to allow their use on other types of data sets. The background, general function, analyses, input, and output are discussed, and instructions for the use of the jobstream are given.

I. BACKGROUND

Many effects of climate on tree growth are not instantaneous. Some effects lag one or more years behind the climatic occurrence affecting growth by a decreasing amount in subsequent growing seasons. After a few years, k , there is no measurable effect of the climate of year t on the growth during year $t+k$. Climate can also influence many internal conditions of the tree and can alter the efficiency of its biological processes in subsequent seasons so that, in addition to climatic lagged effects, wide rings tend to follow wide rings and narrow rings follow narrow ones. In other words, there is positive autocorrelation beyond that caused by lags in the response to climate. Consequently the growth of the tree in the prior year ($t-1$) is also an important modeling component. Both the lagging and autocorrelation relationships must be taken into account to obtain the best estimates of climatic fluctuations.

It has been shown by Fritts (1976) that trees of different species or on different sites often respond differently to the same climatic conditions. Factors such as temperature, precipitation, sunshine, and wind can be directly related

(positively correlated) to ring width in one season on one tree site, but inversely related (negatively correlated) in other seasons and at other tree sites. In addition to regional similarities which allow crossdating, there is some diversity of response which is accentuated by variations in species, topography, and geography which can modify the particular climatic factors that limit growth. Unfavorable effects on growth of one factor in winter sometimes can be offset by favorable effects of the same or other factors in spring or summer. The same extreme condition in one season may limit growth, while in another season, it may favor it.

Since there is a possibility for variation in the growth response from chronology to chronology, the simplistic deductive methods used in earlier years involving inferences drawn from plots or maps of tree-ring chronologies over space (Fritts, 1965) are now viewed as only first approximations of past climatic variations. Far more information can be extracted using linear multivariate calibration methods. Transfer functions with large numbers of coefficients not only accommodate the complexity and diversity of the growth response, but they can also extract seasonal information from differences in the growth response of the various tree samples (Fritts et al., 1971; Fritts, 1976; and Blasing, 1978).

Calibration of a multivariate statistical model to reconstruct a set of predictand variables, such as climate, from a set of predictor variables, such as tree rings, requires several separate computer programs. To reduce the chances for error and to simplify and speed up the running of these programs, they have been grouped into a single package called REPACK (REconstruction PACKage). This manual explains the use of REPACK and discusses some of the mathematics involved.

II. OVERVIEW

This section introduces and briefly describes the programs included in REPACK. The equations involved are presented in Appendix A.

The first program in REPACK is program INPUT which is a logistical program to implement data flow throughout the entire sequence of programs. Program INPUT performs no analyses. (See Figure 1 in Appendix B for a discussion of data flow.)

The first analysis program is BMD06M which calculates the canonical correlations between the predictor variables and the predictand variables. The second analysis program, CANTB, uses the canonical correlations to form a multivariate regression equation as described by Glahn (1968).

Program RECREV uses the multivariate regression equation from program CANTB to estimate past values of the predictand variables. If these variables are eigenvector amplitudes, it transforms these amplitude estimates into estimates of the original predictand by multiplying the amplitude estimates by their corresponding predictand eigenvectors, and computes certain statistics to evaluate the predictand reconstructions in terms of their goodness of fit to the actual data during the calibration period. Some of these statistics (e.g., the percentage of predictand variance calibrated at each point on the grid) can then be printed out in map form via a mapping program.

III. ANALYSES

A. Canonical Correlation--Program BMD06M

The canonical correlation of two sets of variables is done by program BMD06M. The relevant output from this program is written onto disk and sent to program CANTB.

B. Canonical Regression--Program CANTB

Canonical regression is used to obtain transfer function coefficients. The procedure has been modified from the original descriptions to use a stepwise

process which enters each canonical variable (taking the highest canonical correlation first), tests whether the variance due to regression at that particular step is greater than that expected by chance, and accepts or rejects the canonical variable depending on the outcome of an F-test. A weighted or unweighted F-test can be specified (see Variable FWEIGHT in section IV.B). If a variable is rejected and a new variable is entered and accepted at the next or later step, the stepwise procedure reenters each previously rejected variable, one at a time, and accepts any that become significant. This forward and backward checking procedure continues until the least well correlated canonical variate (the last one) has been entered (and accepted or rejected). The coefficients for the canonical regression formed at the highest (or last) *statistically significant* step are selected and provided as input to program RECREV. If no significant regression equation is found, REPACK terminates at this point.

Program CANTB also computes the first-order serial correlation of all input variables (predictor *and* predictand) as well as the serial correlation of the estimated predictand values. The first-order autocorrelation (r_1) of the residuals is calculated at each step, and the residual (denominator) degrees of freedom available for the significance test (dfr) are adjusted to dfr' :

$$dfr' = dfr \frac{1 - r_1}{1 + r_1}$$

If the predictand variables are eigenvector amplitudes, CANTB also computes the percentage of variance calibrated for each predictand eigenvector amplitude, and multiplies it by the fraction of predictand variance reduced by the corresponding eigenvector. The sum of these products is the percentage of the total predictand variance calibrated.

C. Predictand Reconstructions and Calibration Statistics--Program RECREV

This program uses the equation selected in CANTB to reconstruct predictand variables over the period specified by the user. This period can include all of the calibration period and can extend back in time as far as predictor variables are specified.

If the predictand variables are eigenvector amplitudes, estimates of the predictand amplitudes are transformed into "real world" predictand variables by multiplying the reconstructed predictand eigenvector amplitudes by their corresponding eigenvectors. All predictand reconstructions are written onto disk and may later be transferred to a magnetic tape.

Program RECREV then compares the actual predictand data with the reconstructed values in the calibration period by computing the following:

- C1. standard deviations of the residuals at each point on the predictand grid for the calibration period;
- C2. the reduction of error statistic (defined in Appendix A) for the calibration period at each point on the grid (this should be very close to the percentage of predictand variance calibrated);
- C3. the percentage of total predictand variance calibrated at each point on the grid;
- C4. the mean residual at each point on the grid; these should be close to zero (usually between -0.1 and +0.1).

These statistics can be sent to a mapping program for mapping. They are also written onto disk.

RECREV also compares each reconstructed map in the calibration period with the corresponding actual map, thus testing the reconstructions on a year-by-year basis. The following statistics are computed for this purpose:

- C5. the correlation coefficient between each actual map and its corresponding reconstructed map;
- C6. the square of each of the above correlation coefficients;
- C7. the reduction of error statistic between each reconstructed map and its corresponding actual map in the calibration period.

D. Mapped Printout of Reconstructions and of Certain Calibration Statistics

If the predictand variables are in map form, a mapping program can receive the calibration statistics (items C1 to C4 above) from program RECREV and print each out on a base map. If the variable MAPS on the second problem Card (IV.B) is "YES", this program should also print the reconstructed maps for each year in the reconstruction period.

The mapping program, if used, reads unit 9, which is written by program RECREV. This unit contains one header line. If variable CLIVAR is set to P, this header line contains the number of maps on unit 9 (the number years reconstructed, plus four for the four maps of calibration statistics), the initial year of the reconstructions, the season number, the last year of the reconstructions, variable MAPS (see IV.B), and variable MODEL (see IV.B) in (4I5, A5, A9) format. If CLIVAR is not P, this header line contains the number of years reconstructed, the initial year of the reconstructions, the last year of the reconstructions, MAPS, MODEL, and the season name in (3I5, A5, A9, A10) format.

After the header line unit 9 contains the reconstructions, year-by-year, in (96F10.5) format, followed by the four calibration statistics discussed in section III.C in (96F10.5) format.

IV. CARD INPUT

Program INPUT, the first program in REPACK, reads input data from disk files and cards and writes it onto disk files for use by other programs in the jobstream. A data flow chart for the entire sequence of programs is

diagrammed and discussed in Appendix B. INPUT reads the following cards:

A. Problem Card 1 (A10,8I5)

<u>columns</u>	<u>var. name</u>	<u>Description</u>
1-10	PROB	name of run (A10)
11-15	IYRTOR	initial year of predictor variables (I5)
16-20	LSTYTOR	last year of predictor variables (I5)
21-25	IYRCON	initial year of reconstructions (I5)
26-30	LASTYRC	last year of reconstructions (I5)
31-35	IYRAND	initial year of predictand data (I5)
36-40	LSTYAND	last year of predictand data (I5)
41-45	IYCALBR	initial year of the calibration period (I5)
46-50	LSTYCLB	last year of calibration period (I5)

B. Problem Card 2 (4I5,A5,1X,A9,I5,F5.2,I5,2A5,4X,A1)

1-5	NPGRID	number of raw predictand variables (before eigenvector analysis, if applicable) (I5)
6-10	NTOR	number of predictor variables to be used in calibration and reconstruction (I5)
11-15	NTAND	number of predictand variables to be used in calibration (I5)
16-20	NSEASON	number indicating season being reconstructed (I5) 1=summer (Jul-Aug); 2=autumn (Sep-Nov); 3=winter (Dec-Feb); 4=spring (Mar-Apr); 5=spring (May-Jun); 6=spring (Mar-Jun); 7=spring (Apr-Jun); 8=Roberts (Feb-Apr); 9=summer (Jul-Oct); 10=winter (Nov-Feb)
21-25	MAPS	YES or NO: The answer to the question, "Are the reconstructed maps for each year to be printed out?" Must be right-justified (A5)
27-35	MODEL	model being run, e.g., 10p10i10f (A9)
40	MORDER	a number indicating the order in which predictand and predictor data are read in (I1)

- 1 indicates both predictand and predictor are read in for 1 year (all variables) then for the next year, etc. In this option the canonical regression coefficients for all steps (from CANTB) are NOT punched out.
- 2 indicates the predictand is read in for 1 variable (all years) then for the next variable, etc.; predictor is read in as for 1. In this option the canonical regression coefficients ARE punched out.

<u>columns</u>	<u>var. name</u>	<u>description</u>
		3 indicates predictand is read in as for 1; predictor is read in 1 variable (all years) then the next variable (all years) etc. Canonical regression coefficients are punched.
		4 indicates both predictand and predictor are read in 1 variable (all years) then the next variable (all years), etc. Canonical regression coefficients are punched.
		5 same as 1 but canonical regression coefficients are punched.
41-45	FSIG	significant F value for the model being run. Table I indicates the significant F values for various numbers of predictors. The default value is 2.00. (F5.2)
46-50	NOCHRON	number of raw predictor variables (before eigenvector analysis, if applicable). (I5)
51-55	EIGAMPS	Yes or No: The answer to the question, "Are the predictand variables eigenvector amplitudes?" Must be right-justified. (A5)
60	FWIGHT	U if unweighted F-test is desired. W otherwise. Must be right-justified. (A1)
65	CLIVAR	The climatic variable: P for pressure T for temperature R for precipitation D for drought indices H for 700 mb heights

If CLIVAR is P, the predictand eigenvectors, if needed, are read with a READ statement which alters their positions in the array, a five-line header is assumed to exist on unit 48, and the actual predictand data is read with a double implied DO loop [READ(...)((ACT(I,J), J=1, NPGRID), I=1, NYTAND)]. If CLIVAR is anything other than P, it is assumed that the predictand eigenvectors are in the correct order, no header is assumed on unit 48, and the actual predictand data is read with an actual DO loop and a single implied DO loop

```
[DO 121 I = 1, NYTAND
  READ(...) (ACT (I,J), J=1, NPGRID)
121 CONTINUE] . (A1)
```

- C. A card indicating which predictand variables the model uses (e.g., to use variables 1, 2, and 5, type 1 2 5) (20I4).
- D. Format for predictand data (8A10)
- E. Format for predictor data. (8A10/8A10) There must be 2 cards even if the second one is blank.
- F. Format for actual predictand data (8A10)
- G. One blank card

TABLE I

Significant F Level to Be Used in REPACK
for Five to 30 Predictor Variables. All
Values Were Interpolated by T. J. Blasing from
The Tables of Burington and May

Number of Predictor Variables Used	Significant F Level
5	2.25
6	2.13
7	2.04
8	1.97
9	1.91
10	1.86
11	1.82
12	1.79
13	1.75
14	1.71
15	1.70
16	1.68
17	1.67
18	1.65
19	1.63
20	1.61
21	1.60
22	1.59
23	1.58
24	1.56
25	1.56
26	1.55
27	1.54
28	1.53
29	1.52
30	1.51

V. RUNNING INSTRUCTIONS

It is suggested that users read this entire manual before running REPACK.

Prior to running REPACK much of the data must be put on disk files.

The predictand variables should be on unit 31. These variables may be in any format because the user specifies the predictand format (see IV.D).

The predictor variables should be on unit 32. These variables may be in any format because the user specifies the predictor format (see IV.E).

The predictand eigenvectors, if needed, should be on unit 33. These variables must be in (10X, 10F7.3) format.

The percentage of the predictand variance reduced by each predictand variable, if needed, should be on unit 34. These variables must be in (10X, 10F7.3) format.

The actual predictand variables, before the transformation by eigenvector analysis, if applicable, should be on unit 48. If CLIVAR (see IV.B) is P, five header lines, containing the season number (see IV.B), the initial year, last year, and number of years of the actual predictand variables, and an alphanumeric description of these variables, should be on this unit in (I2, 2I5, I3, A6, 6A10, /, 3(8A10, /), 8A10) format. If CLIVAR is not P, header lines should NOT be on this file. The actual predictand variables may be in any format because the user specifies the actual predictand format (see V.F).

Once the data has been properly located, the programs should be executed beginning with program INPUT, then programs BMDO6M, CANTB, RECREV, and finally the mapping program, if needed.

It is suggested that the computer preset all locations to a value which it is illegal to use in a mathematical operation such as addition, if this is possible. Line number 10 in program INPUT sets a variable explicitly to such a value for a CDC computer. Line number 297 in program CANTB uses the CDC function LEGVAR to test whether this variable has had another value assigned to it.

LEGVAR returns a value of zero if the tested variable has a legal value; it returns a non-zero value otherwise. The variable in question will have an illegal value if the predictand variables are not eigenvector amplitudes. When the predictand variables are not eigenvector amplitudes, the percentage of the total predictand variance calibrated (VI.A.3.k.iii) is output as zero, and the number output as the percentage of the total predictand amplitude variance calibrated is really the percentage of the total predictand variance calibrated.

After all programs have executed, unit 51 will contain the calibration statistics C1. through C4. (see III.C) and unit 50 will contain the reconstructions, year by year. These files should be saved as the user desires.

VI. OUTPUT

A. Printed Output

1. From program INPUT
 - a. Name of run, date and time of execution, model description
 - b. List of predictand data used in calibration
 - c. List of predictor data used in calibration
 - d. First and last years of predictor data used in reconstruction
2. From program BMD06M
 - a. Input data for both predictor and predictand variables
 - b. Standard deviations of all input variables
 - c. Correlation matrix of all input variables
 - d. Canonical correlations and "coefficients"
 - e. Canonical check matrix
3. From program CANTB
 - a. X (data) (predictor data)
 - b. Matrix B (same as "coefficients" in 2-d above; "coefficients" corresponding to the predictand form matrix B.)
 - c. Matrix A (same as "coefficients" in 2-d above; "coefficients" corresponding to the predictor form matrix A.)
 - d. Lambdas (canonical correlations)
 - e. Y (data) (predictand data)
 - f. Covariance submatrix of Y (correlation matrix of the predictand variables)
 - g. Means and standard deviations of the X variables
 - h. Means and standard deviations of the Y variables
 - i. (Canonical) amplitudes of X
 - j. (Canonical) amplitudes of Y

- k. The following are output each time a λ is added to regression:
 - i. R squared, adjusted R squared, and percentage of predictand variance calibrated (if applicable) for each predictand variable in the model;
 - ii. average adjusted R squared (over all predictand variables) and its increase since the last step;
 - iii. percentage of total predictand variance calibrated (if applicable) and percentage of total predictand amplitude variance calibrated;
 - iv. ANOVA table for significance testing
- l. YHTSV (estimated values of the predictand obtained using the canonical regression equation selected by F-testing)
- m. BETASV (B's used to estimate YHTSV from predictor data)
- n. Number of the last significant canonical correlation used in the model
- o. First-order serial correlation coefficients of predictor, predictand, and estimated values of predictand variables

4. From program RECREV

- a. First and last predictand eigenvectors used, if applicable
- b. Beta's used (BETASV) as a check on correct transmission from CANTB
- c. Reconstructed predictand variables
- d. First and last year's actual predictand data in the dependent portion of the reconstructions
- e. Yearly correlation coefficients between reconstructed and actual maps for the calibration period, the correlation coefficients squared, and the reduction of error statistic for each year in the calibration period
- f. Percent variance calibrated by the model, and counts of the number of map correlations and reduction of errors (r and RE) greater than or equal to certain values.

B. Disk File Output from Program RECREV

- 1. Standard deviation of the residuals at each point
- 2. Reduction of error at each point
- 3. Calibrated variance at each point
- 4. Mean residual at each point
(Items 1-4 go on unit 51)
- 5. (On unit 50) Reconstructed predictand variables at all points for all years reconstructed (year by year)

C. Punched Output

- 1. From program CANTB
 - a. Means and standard deviations of the x(predictor) data
 - b. Means and standard deviations of the y(predictand) data
 - c. If Morder > 1, betas for each canonical correlation
- 2. From Program RECREV, a card containing the season number, CLIVAR, number of raw predictand variables NPGRID (see V.B), the digit

one, the number of years reconstructed for which there is actual predictand data, MODEL, the percent variance calibrated, and the counts of the number of correlation coefficients and reduction of error values greater than or equal to certain values.

APPENDIX A

SUMMARY OF EQUATIONS USED

A. BMD06M--Canonical Correlation

Given n observations of p predictor variables, ${}_nX_p$, and n corresponding observations of q predictand variables ${}_nY_q$, the following correlation matrices are defined:

$$S_{11} = \frac{1}{n} X^t X \quad S_{22} = \frac{1}{n} Y^t Y \quad S_{12} = \frac{1}{n} X^t Y \quad S_{21} = \frac{1}{n} Y^t X$$

where the superscript t denotes the transpose.

The program then finds eigenvectors, A , corresponding to the predictor variables; eigenvectors, B , corresponding to the predictand variables; and the canonical correlations, λ_i , $i=1, \dots, r$ (where r is the number of possible canonical correlations) which satisfy the following equations.

$$(S_{11}^{-1} S_{12} S_{22}^{-1} S_{21} - \lambda_i^2 I) A_i = 0 \quad (S_{22}^{-1} S_{21} S_{11}^{-1} S_{12} - \lambda_i^2 I) B_i = 0 \quad i=1, r$$

B. CANTB--Canonical Regression

This program computes the r sets of possible β 's for the canonical regression equation $\hat{Y} = X\beta$

where $\hat{Y}$ is the matrix of reconstructed pressure eigenvector amplitudes and β is found as follows: $\beta = A\Lambda B S_{22}^{-1}$

Λ is a diagonal matrix each element of which is either the canonical correlation of a significant step or zero if the step is not significant.

Y is compared with $\hat{Y}$ at each step to obtain the percentage of Y variance calibrated. This is R^2 (times 100%) where $R^2 = \frac{1}{q} \sum_{j=1}^q R_j^2$

q is the number of predictand variables, and $R_j^2 = R^2$ for the jth Y variable =

$$\frac{\left[\sum_{i=1}^n (Y_{j,i} - \bar{Y}_j) (\hat{Y}_{j,i} - \bar{\hat{Y}}_j) \right]^2}{\sum_{i=1}^n (Y_{j,i} - \bar{Y}_j)^2 \sum_{i=1}^n (\hat{Y}_{j,i} - \bar{\hat{Y}}_j)^2}$$

This value is then partially adjusted for degrees of freedom using $R_a^2 = \text{adjusted } R^2$
 $= 1 - (1 - R^2) \left(\frac{nq - q}{nq - pk} \right)$

where p is the number of predictor variables, q is the number of predictand variables, and this computation is made at a step including k variates in the determination of β .

At each step (each time a λ_i is added as a diagonal element of Λ) a sequential F-test is performed as follows:

$$\text{F-ratio} = \frac{\frac{\hat{SSY}_{\text{inc}}}{p}}{\frac{SSR}{\left(\frac{1-r_s}{1+r_s} \right) (nq-pk)}}$$

where $\hat{SSY}_{\text{inc}}$ is the sum of squares increase due to regression since the last preceding significant step, r_s is the first-order serial correlation of the residual ("unexplained") portion of the variance, and p, q, and k have the same meaning as above. If the weighted F-test has been specified, both the standardized Y and $\hat{Y}$ variables are multiplied by their standard deviations before the F-ratio is computed.

C. RECREV--Evaluation of the calibration

The residual predictand ϵ_{ij} is defined as the actual value minus the reconstructed value for each year i in the calibration period at the jth gridpoint. The standard deviation of these residuals at the jth gridpoint, SR_j , is computed

$$\text{as } \overline{\epsilon_j} = \frac{\sum_{i=1}^n \epsilon_{i,j}^2 - \frac{(\sum_{i=1}^n \epsilon_{i,j})^2}{n}}{n-1}$$

where $\overline{\epsilon_j}$ is the mean residual at the jth gridpoint and n is the number of years in the calibration period. This provides estimates of the "expected" error term at each gridpoint.

The reduction of error statistic, R. E., is expressed as $R. E. = 1 - \frac{SSR}{SSM}$ where SSR is the sum of squares of the residuals and SSM is the sum of squares of departures of the predictand from its mean value over the calibration period.

APPENDIX B

DATA FLOW

A. Program INPUT does the following:

1. Identification of the run, information on which predictand variables to use, predictand variables, and predictor variables are written on a disk for use in program BMD06M.
2. Identification of the run, information on the order of the predictor and predictand data, which set of betas to use, and which variables to use; the percent of predictand variance reduced by the predictand eigenvectors used (if applicable); predictand variables, and predictor variables are written on a disk for use by program CANTB.
3. Information on which predictand eigenvectors to use (if applicable), identification of the run, predictand eigenvectors (if applicable), and predictor variables are written on a disk for use by program RECREV.
4. Selected input data is printed out to provide a check on the input data.

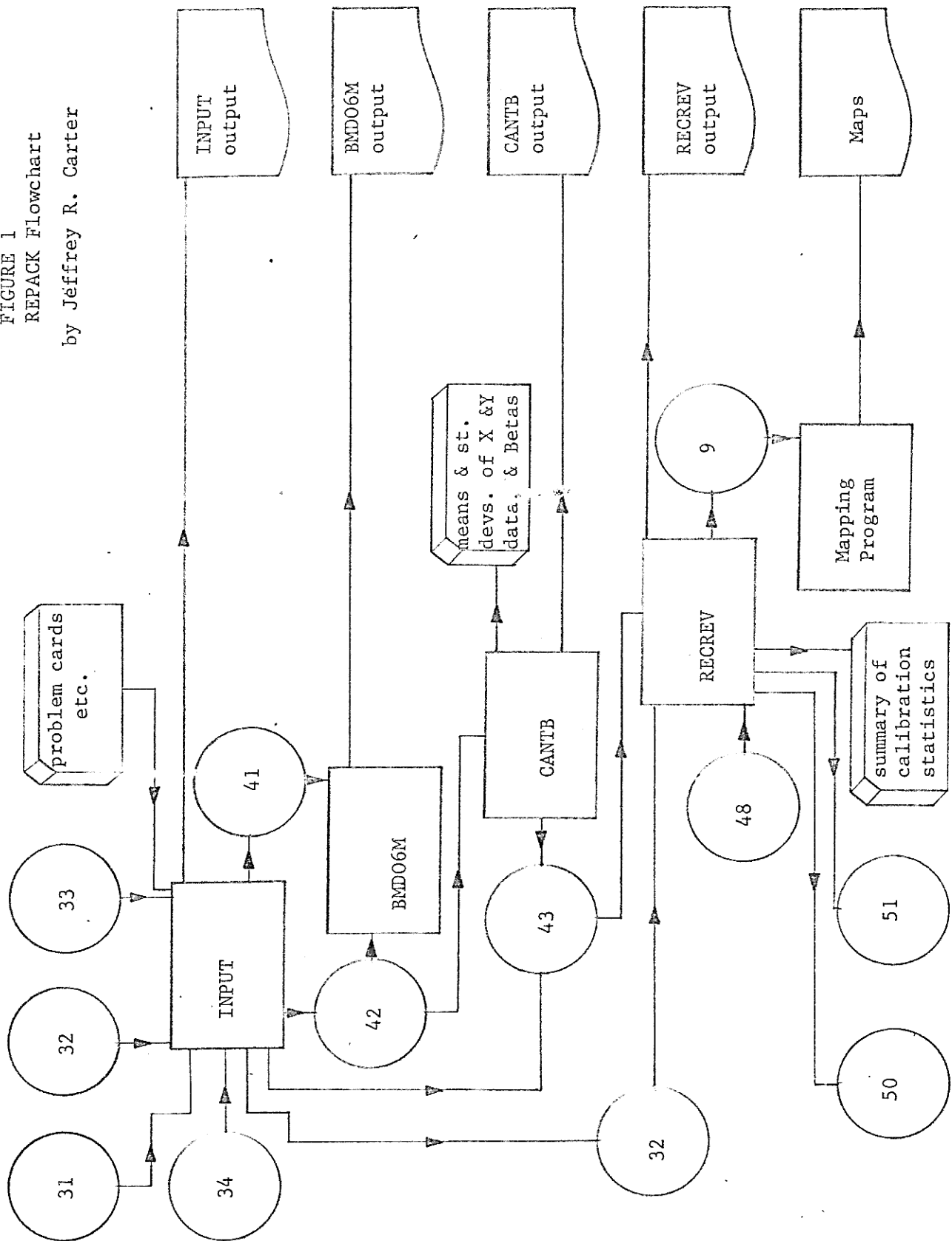
B. Other programs generate data for use in succeeding programs, or to be output on paper, disk files, or cards. A data flow chart is on the next page.

APPENDIX C

IMPLEMENTATION AT THE LABORATORY OF TREE-RING RESEARCH

REPACK is run at the Laboratory of Tree-Ring Research on a CDC Cyber 175 computer operating under SCOPE 3.4.4. REPACK is used to reconstruct climate from tree-ring indices. Eigenvectors of both the predictand and the predictor are used to reduce the number of variables being used. Sea-level pressure over western North America and the North Pacific Sector, temperature over the continental United States, and precipitation over the continental United States are the climatic variables reconstructed.

FIGURE 1
 REPACK Flowchart
 by Jeffrey R. Carter



The procedure for a pressure reconstruction differs slightly from that for temperature and precipitation reconstructions. The map for the pressure grid covers a different area than that for temperature and precipitation, and the pressure values are distributed on a rectangular grid, which is not the case for temperature and precipitation. For these reasons different mapping routines are used for the two different maps. Two additional programs, RDPRES and TYPCOR, are run with REPACK for pressure but for no other predictand, and a program called PVEEST is run for temperature and precipitation but not for pressure.

REPACK was originally written to reconstruct pressure, and certain modifications remain in program INPUT to facilitate the running of programs RDPRES and TYPCOR. These are brought into play only when CLIVAR is P and involve writing some information onto disk files which are otherwise unused. The blank card of Section IV.G is used in this case to read in information for TYPCOR.

The actual pressure data are on a tape in unformatted (binary) form for all seasons. Program RDPRES reads the season number, searches out the data for the desired season, and writes it on unit 48 in a format which RECREV can use. In the process RDPRES puts a five line header on the data.

Using the map pattern correlation process described by Blasing (1975), pressure anomaly type patterns have been obtained for each season. Program TYPCOR correlates every year of reconstructed pressure with these patterns. It then compares the years reconstructed as being of a type with those years which actually are a type during the calibration period to provide another indication of the quality of the calibration.

Because the pressure data is on a regular rectangular grid, a simplified mapping routine is used which avoids redundant operations which would take place if the temperature and precipitation routine were used.

The deck set-up for an autumn pressure calibration run using the CDC SCOPE operating system at the University of Arizona is:

```
PREPACK, BNxxxxxxxx,CM130K,T40,P2,ST76.
```

```
ATTACH(PREPACKAUTUMNDATA,ID=JHH)
```

This permanent disk file contains pressure data and input for TYPCOR. The TYPCOR input will be transferred to unit 35 (TAPE35).

```
COPYBF(PREPACK,TAPE31)
COPYBF(PREPACK,TAPE33)
COPYBF(PREPACK,TAPE34)
COPYBF(PREPACK,TAPE35)
REWIND(TAPE31,TAPE33,TAPE34,TAPE35)
ATTACH(TAPE95,ID=JHH)
```

This file contains information for RDPRES (TAPE1) and the mapping program (TAPE71), the tree-ring data, and the binary object codes of the programs.

```
SKIPF(TAPE95,2,17B)
COPYBF(TAPE95,TAPE71)
COPYBF(TAPE95,TAPE1,14)
COPYBF(TAPE95,INPUT)          binary deck of INPUT called INPUT
                                because INPUT is a reserved name

COPYBF(TAPE95,BMD06M)
COPYBF(TAPE95,BMDSUBR)        subroutines used by BMD06M
COPYBF(TAPE95,CANTB)          binary deck of CANTB
COPYBF(TAPE95,RECREV)         binary deck of RECREV
COPYBF(TAPE95,PRESRCN)        binary deck of the mapping program
COPYBF(TAPE95,TYPCOR)         binary deck of TYPCOR
COPYBF(TAPE95,SDPRES)         binary deck of RDPRES
SKIPF(TAPE95,7,17,B)
COPYBF(TAPE95,TAPE32)
REWIND(TAPE32,TAPE71,TAPE1)
LDSET(PRESET=NGINDEF)         presets computer to illegal values
INPUT.                         loads and executes INPUT
LDSET(PRESET=NGINDEF)
LOAD(BMD06M)
LOAD(BMDSUBR)
EXECUTE.
LDSET(PRESET=NGINDEF)
CANTB.
LDSET(PRESET=NGINDEF)
RDPRES.                        prepares unit 48 for RECREV
LDSET(PRESET=NGINDEF)
RECREV.
LDSET(PRESET=NGINDEF)
PRESRCN.
LDSET(PRESET=NGINDEF)
TYPCOR.
REWIND(TAPE50,TAPE51)
ATTACH(SVFILE,ID=JHH,PW=xxxxx)
REK(SVFILE/R)                 sets SVFILE ready to copy onto
COPYBF(TAPE51,SVFILE)          saves disk file output from RECREV
COPYBF(TAPE50,SVFILE)
EXTEND(SVFILE)                 extends the length of SVFILE
```

THE CARDS DESCRIBED IN SECTION IV GO HERE.
 CLIVAR is P.
 7/8/9
 6/7/8/9

The mapping of temperature and precipitation is done with subroutine PRINTMP, a highly versatile mapping routine capable of cheaply and quickly producing almost any map. Information on subroutine PRINTMP can be obtained from the Laboratory of Tree-Ring Research. The maps are also being contoured by computer using subroutine CONTUR2.

The program which computes eigenvectors lists the percent variance reduced by each eigenvector but does not punch them. For pressure these values have been hand punched and transferred to disk, but for temperature and precipitation this has not been done. For this reason program PVEEST is run prior to the execution of INPUT. PVEEST estimates the percent variance reduced by each of the first twenty eigenvectors of the predictand.

The deck set-up for a temperature or precipitation run is similar to that of a pressure run except:

The jobname is TREPACK
 PVEEST is run prior to INPUT
 RPPPC is not run and no input data is obtained for it
 A different mapping program is used
 TYPCOR is not run and no input data is obtained for it
 CLIVAR is T or R.

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